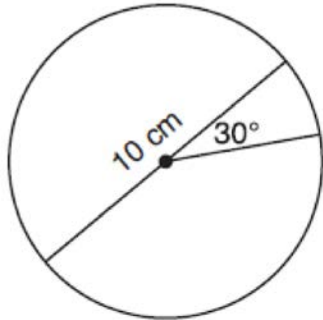


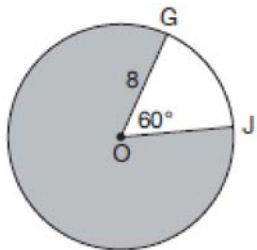
G.C.B.5: Sectors

- 1 A circle with a diameter of 10 cm and a central angle of 30° is drawn below.



What is the area, to the *nearest tenth of a square centimeter*, of the sector formed by the 30° angle?

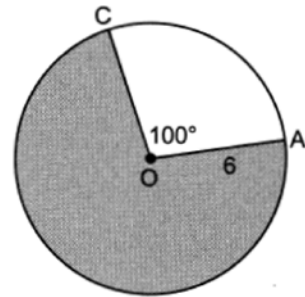
- 1) 5.2
 - 2) 6.5
 - 3) 13.1
 - 4) 26.2
- 2 In the diagram below of circle O , $GO = 8$ and $m\angle GOJ = 60^\circ$.



What is the area, in terms of π , of the shaded region?

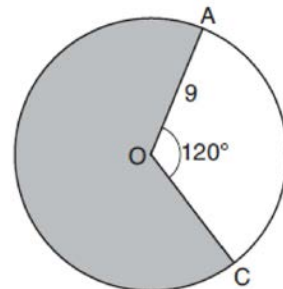
- 1) $\frac{4\pi}{3}$
- 2) $\frac{20\pi}{3}$
- 3) $\frac{32\pi}{3}$
- 4) $\frac{160\pi}{3}$

- 3 In circle O below, $OA = 6$, and $m\angle COA = 100^\circ$.



What is the area of the shaded sector?

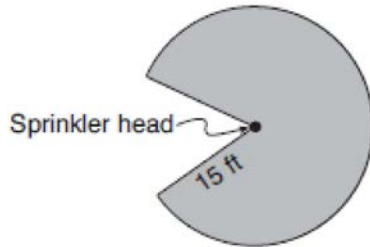
- 1) 10π
 - 2) 26π
 - 3) $\frac{10\pi}{3}$
 - 4) $\frac{26\pi}{3}$
- 4 Circle O with a radius of 9 is drawn below. The measure of central angle AOC is 120° .



What is the area of the shaded sector of circle O ?

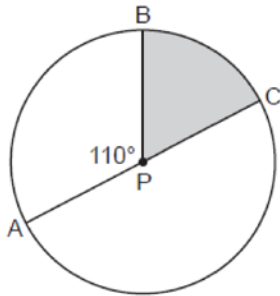
- 1) 6π
- 2) 12π
- 3) 27π
- 4) 54π

- 5 Cerise waters her lawn with a sprinkler that sprays water in a circular pattern at a distance of 15 feet from the sprinkler. The sprinkler head rotates through an angle of 300° , as shown by the shaded area in the accompanying diagram.



What is the area of the lawn, to the *nearest square foot*, that receives water from this sprinkler?

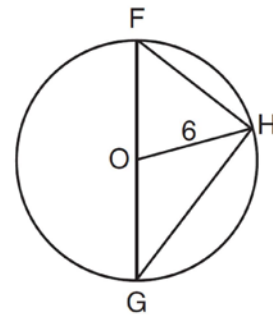
- 1) 79
 - 2) 94
 - 3) 589
 - 4) 707
- 6 In circle P below, diameter $\overline{AC}$ and radius $\overline{BP}$ are drawn such that $m\angle APB = 110^\circ$.



If $AC = 12$, what is the area of shaded sector BPC ?

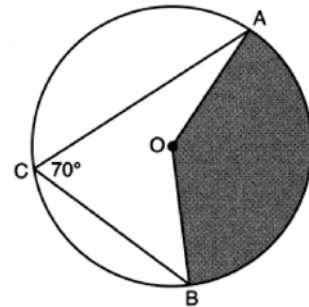
- 1) $\frac{7}{6}\pi$
- 2) 7π
- 3) 11π
- 4) 28π

- 7 Triangle FGH is inscribed in circle O , the length of radius $\overline{OH}$ is 6, and $\overline{FH} \cong \overline{OG}$.



What is the area of the sector formed by angle FOH ?

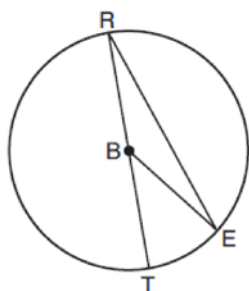
- 1) 2π
 - 2) $\frac{3}{2}\pi$
 - 3) 6π
 - 4) 24π
- 8 In the diagram below of circle O , $\overline{AC}$ and $\overline{BC}$ are chords, and $m\angle ACB = 70^\circ$.



If $OA = 9$, the area of the shaded sector AOB is

- 1) 3.5π
- 2) 7π
- 3) 15.75π
- 4) 31.5π

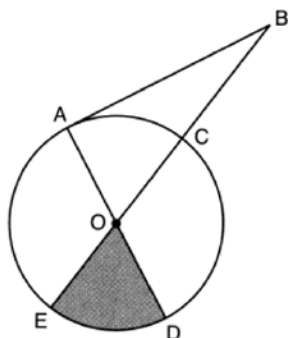
- 9 In circle B below, diameter $\overline{RT}$, radius $\overline{BE}$, and chord $\overline{RE}$ are drawn.



If $m\angle TRE = 15^\circ$ and $BE = 9$, then the area of sector EBR is

- 1) 3.375π
- 2) 6.75π
- 3) 33.75π
- 4) 37.125π

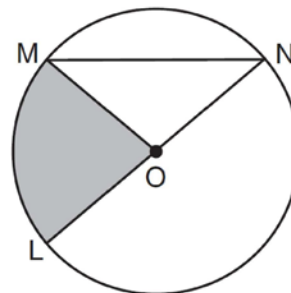
- 10 In the diagram below of circle O , tangent $\overline{AB}$ is drawn from external point B , and secant $\overline{BCOE}$ and diameter $\overline{AOD}$ are drawn.



If $m\angle OBA = 36^\circ$ and $OC = 10$, what is the area of shaded sector DOE ?

- 1) $\frac{3\pi}{10}$
- 2) 3π
- 3) 10π
- 4) 15π

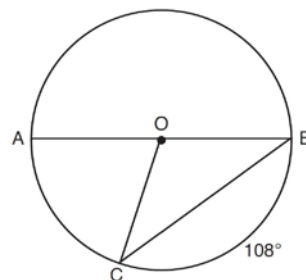
- 11 In the diagram below of circle O , the area of the shaded sector LOM is $2\pi \text{ cm}^2$.



If the length of $\overline{NL}$ is 6 cm, what is $m\angle N$?

- 1) 10°
- 2) 20°
- 3) 40°
- 4) 80°

- 12 In circle O , diameter $\overline{AB}$, chord $\overline{BC}$, and radius $\overline{OC}$ are drawn, and the measure of arc BC is 108° .



Some students wrote these formulas to find the area of sector COB :

Amy $\frac{3}{10} \cdot \pi \cdot (BC)^2$

Beth $\frac{108}{360} \cdot \pi \cdot (OC)^2$

Carl $\frac{3}{10} \cdot \pi \cdot \left(\frac{1}{2}AB\right)^2$

Dex $\frac{108}{360} \cdot \pi \cdot \frac{1}{2}(AB)^2$

Which students wrote correct formulas?

- 1) Amy and Dex
- 2) Beth and Carl
- 3) Carl and Amy
- 4) Dex and Beth

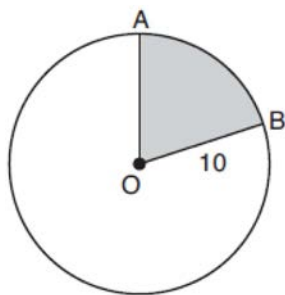
- 13 The area of a sector of a circle with a radius measuring 15 cm is 75π cm². What is the measure of the central angle that forms the sector?

1) 72°
2) 120°
3) 144°
4) 180°

- 14 What is the area of a sector of a circle with a radius of 8 inches and formed by a central angle that measures 60° ?

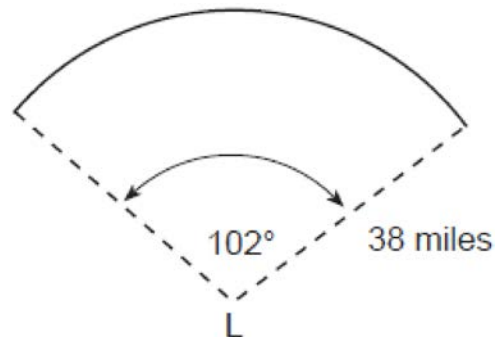
1) $\frac{8\pi}{3}$
2) $\frac{16\pi}{3}$
3) $\frac{32\pi}{3}$
4) $\frac{64\pi}{3}$

- 15 In the diagram below, circle O has a radius of 10.



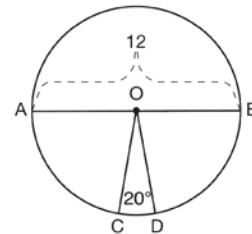
If $m\widehat{AB} = 72^\circ$, find the area of shaded sector AOB , in terms of π .

- 16 The diagram below models the projection of light from a lighthouse, L . The sector has a radius of 38 miles and spans 102° .



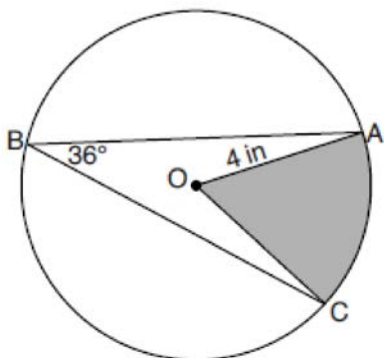
Determine and state the area of the sector, to the nearest square mile.

- 17 In the diagram below of circle O , diameter $\overline{AB}$ and radii $\overline{OC}$ and $\overline{OD}$ are drawn. The length of $\overline{AB}$ is 12 and the measure of $\angle COD$ is 20 degrees.



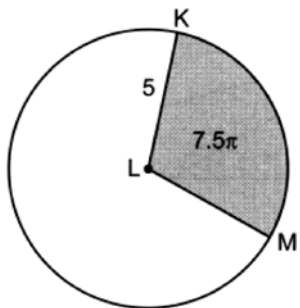
If $\widehat{AC} \cong \widehat{BD}$, find the area of sector BOD in terms of π .

- 18 In the diagram below of circle O , the measure of inscribed angle ABC is 36° and the length of $\overline{OA}$ is 4 inches.



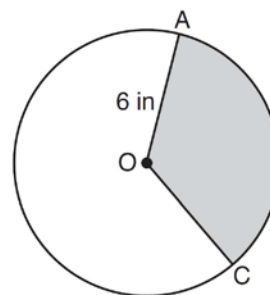
Determine and state, to the *nearest tenth of a square inch*, the area of the shaded sector.

- 19 In the diagram below of circle L , the area of the shaded sector KLM is 7.5π and $LK = 5$.

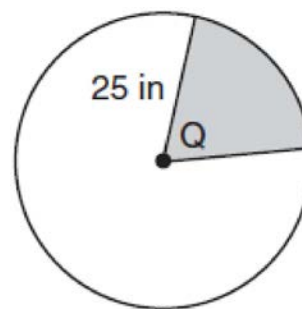


Determine and state the degree measure of angle KLM , the central angle of the shaded sector.

- 20 In the diagram below of circle O , the area of the shaded sector AOC is $12\pi \text{ in}^2$ and the length of $\overline{OA}$ is 6 inches. Determine and state $m\angle AOC$.



- 21 In the diagram below, the circle has a radius of 25 inches. The area of the *unshaded* sector is $500\pi \text{ in}^2$.



Determine and state the degree measure of angle Q , the central angle of the shaded sector.

- 22 A circle has a radius of 6.4 inches. Determine and state, to the *nearest square inch*, the area of a sector whose arc measures 80° .
- 23 Determine and state, in terms of π , the area of a sector that intercepts a 40° arc of a circle with a radius of 4.5.

G.C.B.5: Sectors

Answer Section

1 ANS: 2

$$\frac{30}{360} (5)^2 (\pi) \approx 6.5$$

REF: 081818geo

2 ANS: 4

$$\frac{300}{360} \cdot 8^2 \pi = \frac{160\pi}{3}$$

REF: 011721geo

3 ANS: 2

$$\left(\frac{360 - 100}{360} \right) (\pi) (6^2) = 26\pi$$

REF: 062411geo

4 ANS: 4

$$\left(\frac{360 - 120}{360} \right) (\pi) (9^2) = 54\pi$$

REF: 081912geo

5 ANS: 3

The area of the entire circle is $15^2 \pi = 225\pi$. The shaded area has an area of $225\pi \times \frac{300}{360} \approx 589$

REF: 060716b

6 ANS: 2

$$\frac{70}{360} \cdot 6^2 \pi = 7\pi$$

REF: 082309geo

7 ANS: 3

$$\frac{60}{360} \cdot 6^2 \pi = 6\pi$$

REF: 081518geo

8 ANS: 4

$$\frac{140}{360} \cdot 9^2 \pi = 31.5\pi$$

REF: 012317geo

9 ANS: 3

$$\frac{150}{360} \cdot 9^2 \pi = 33.75\pi$$

REF: 012013geo

10 ANS: 4

$$\frac{54}{360} \cdot 10^2 \pi = 15\pi$$

REF: 062224geo

11 ANS: 3

$$\frac{x}{360} \cdot 3^2 \pi = 2\pi \quad 180 - 80 = 100$$

$$x = 80 \quad \frac{180 - 100}{2} = 40$$

REF: 011612geo

12 ANS: 2 REF: 081619geo

13 ANS: 2

$$\frac{x}{360} (15)^2 \pi = 75\pi$$

$$x = 120$$

REF: 011914geo

14 ANS: 3

$$\frac{60}{360} \cdot 8^2 \pi = \frac{1}{6} \cdot 64\pi = \frac{32\pi}{3}$$

REF: 061624geo

15 ANS:

$$\frac{72}{360} (\pi) (10^2) = 20\pi$$

REF: 061928geo

16 ANS:

$$\frac{102}{360} (\pi) (38^2) \approx 1285$$

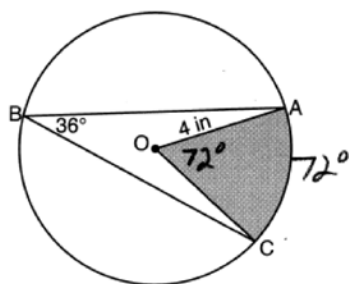
REF: 012426geo

17 ANS:

$$\frac{\left(\frac{180-20}{2}\right)}{360} \times \pi (6)^2 = \frac{80}{360} \times 36\pi = 8\pi$$

REF: spr1410geo

18 ANS:



$$\left(\frac{72}{360}\right)\pi(4)^2 \approx 10.1$$

REF: 082231geo

19 ANS:

$$\frac{7.5\pi}{25\pi} \cdot 360 = 108$$

REF: 082530geo

20 ANS:

$$A = 6^2 \pi = 36\pi \quad 36\pi \cdot \frac{x}{360} = 12\pi$$

$$x = 360 \cdot \frac{12}{36}$$

$$x = 120$$

REF: 061529geo

21 ANS:

$$\frac{Q}{360}(\pi)(25^2) = (\pi)(25^2) - 500\pi$$

$$Q = \frac{125\pi(360)}{625\pi}$$

$$Q = 72$$

REF: 011828geo

22 ANS:

$$\frac{80}{360} \cdot \pi(6.4)^2 \approx 29$$

REF: 062328geo

23 ANS:

$$\frac{40}{360} \cdot \pi(4.5)^2 = 2.25\pi$$

REF: 061726geo